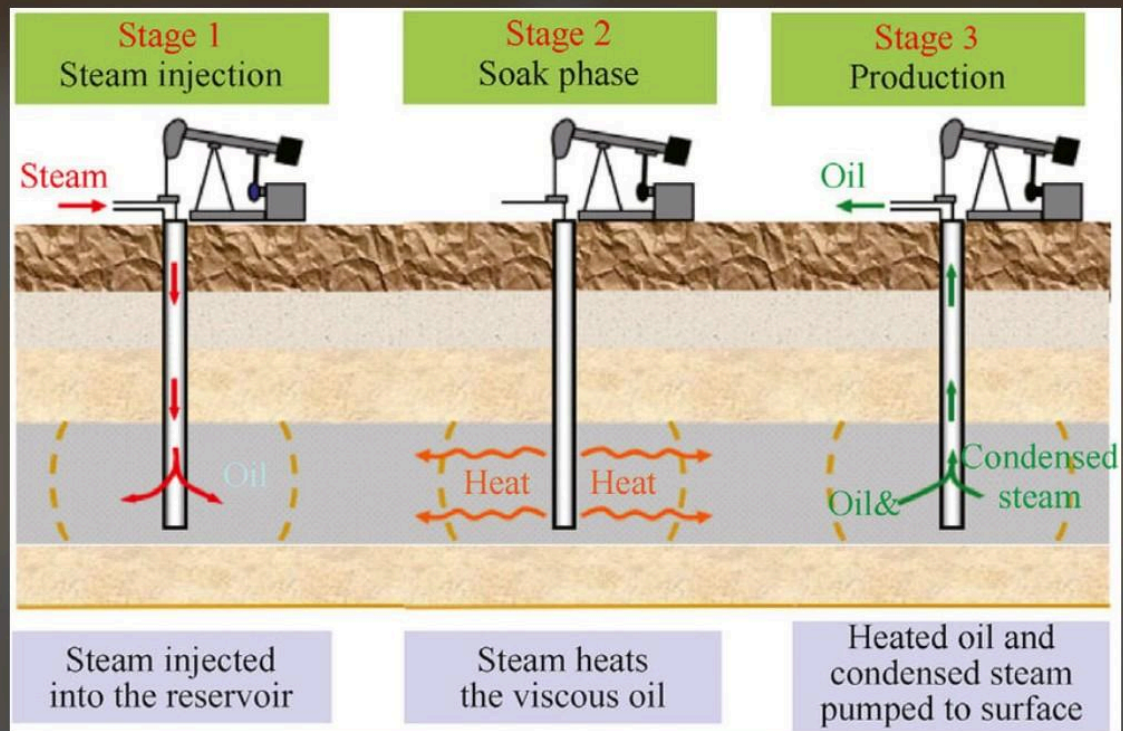


MATHEMATICAL MODELLING (THERMAL+SRP)



SIH $\rightarrow$ (Thermal Modelling)For the thermal part $\rightarrow$ Three eqns. (Basic) $\rightarrow$ (1) Thermo: Steam Enthalpy Egn.

• Calculates the total thermal energy delivered to the wellbore based on mass and quality of steam injected $\rightarrow$

$$Q_{in} = m_s \cdot \{ n \cdot h_{fg} + C_w \cdot (T_{steam} - T_{initial}) \}$$

(2) Reservoir Absorption Egn. $\rightarrow$

• It calculates how much injected thermal energy actually raises the tempⁿ. of rock & oil.

C_{total} (Heat Capa. of heated zone):

$$C_{total} = (m_{rock} \cdot C_{rock}) + (m_{oil} \cdot C_{oil})$$

$$\text{Change in temp. } (\Delta T) = \frac{Q_{in} \cdot \eta}{C_{total}}$$

$$\text{New temp. } (T_{new}) = T_{initial} + \Delta T$$

Today's Priority

March		2010			
Monday	1	8	15	22	29
Tuesday	2	9	16	23	30
Wednesday	3	10	17	24	31
Thursday	4	11	18	25	
Friday	5	12	19	26	
Saturday	6	13	20	27	
Sunday	7	14	21	28	

(3) Fluid Dynamics: Beggs-Robinson

HT2

Correlation →

• To calculate the exact drop in heavy crude oil viscosity based on new temp.

First Convert temp. in (°C) to (F).

$$\checkmark \bullet T_f = (T_{new} \times 1.8) + 32$$

$$\checkmark \bullet Z = 3.0324 - (0.02023 \cdot API)$$

↙
baseline
const

$$Y = 10^Z$$

API = American Petroleum Institute Gravity.
→ (Works in reverse scale).

In problem statement: ~18° API Crude Oil.

• Thermal Decay (X) →

$$X = Y \cdot T_f^{-1.163}$$

↘ with inc. value of
(T_f), (X) gets smaller.

• Final Viscosity (μ) → (In Centipoise
Cp)

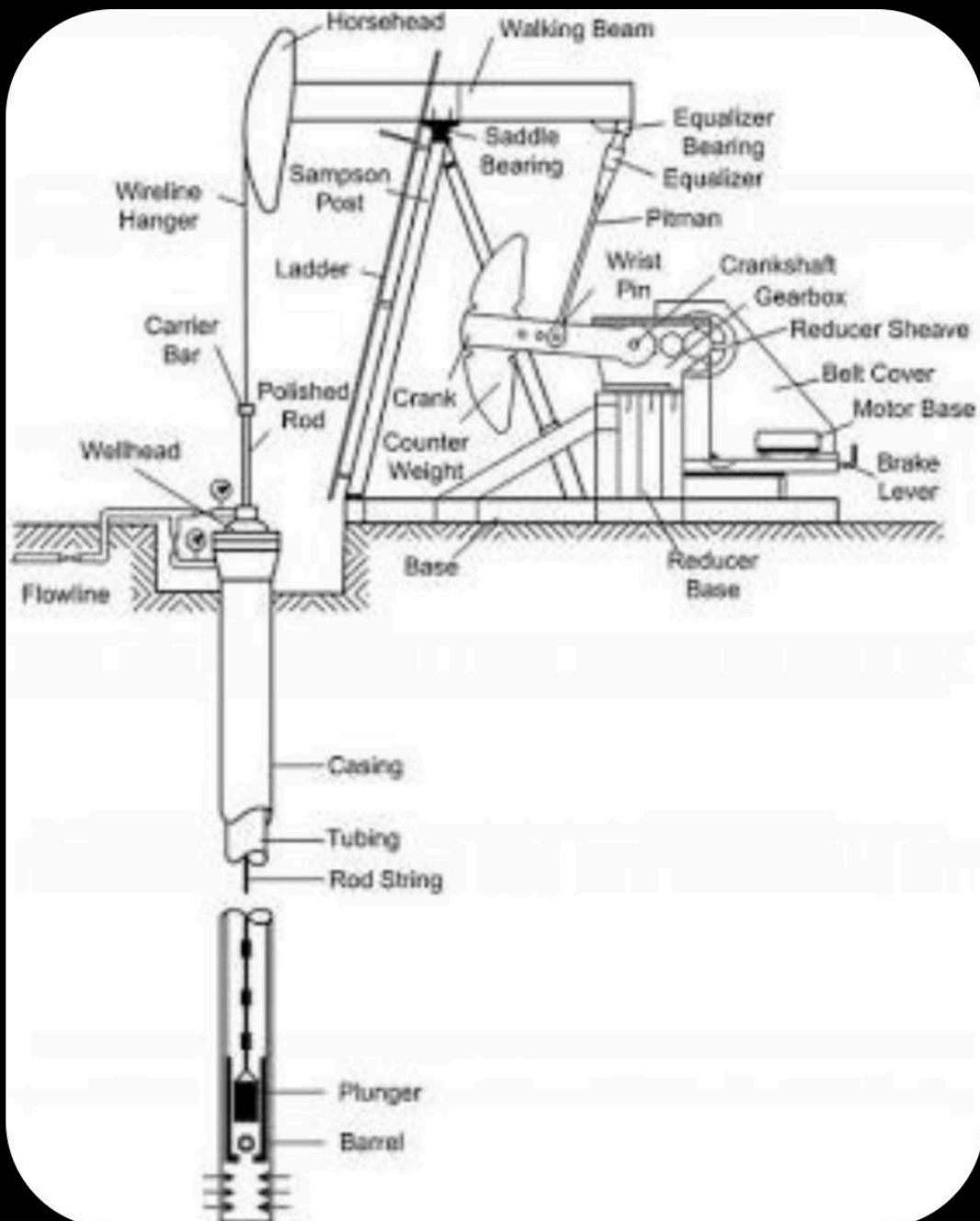
$$\mu = 10^X - 1$$

At initial temp. (Around 47°C), Crude oil has 66 cp (Approx).

At 250°C, it reaches close to 1, meaning successfully dec. viscosity until it flows like water.

February	2010
Monday	1 8 15 22
Tuesday	2 9 16 23
Wednesday	3 10 17 24
Thursday	4 11 18 25
Friday	5 12 19 26
Saturday	6 13 20 27
Sunday	7 14 21 28

Today's Priority



SUCKER ROD EQN:

- To Calculate how to physically lift ~~the~~ without breaking the machinery.

• Rod stretch eqn. $\rightarrow$ (Hooke's Law).

$$\Delta L = \frac{F_{load} \cdot L}{A \cdot E}$$

It calculates how much the metal rod stretches under the weight of oil. (# Sucker rod is a series of steel rods).

• Peak-Polished Rod Load (PPRL) $\rightarrow$

$$PPRL = W_r + W_f + F_{drag}$$

• Calculates the absolute max. tension on the topmost rod just as the pump starts its upward stroke.

W_r = Weight of thousands of metres of steel rods.

W_f = Weight of fluid

F_{drag} = Viscous friction of oil fighting against the rods.

Today's Priority

March 2010	
Monday	1 8 15 22 29
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Why Rod stretch matters??

Why this matters: If the motor on the surface pulls the rod up by 2 meters (the stroke), but the heavy oil causes the rod to stretch by 1.5 meters, the plunger at the bottom of the well only moves 0.5 meters. The stretch "steals" your pumping efficiency.

Your connection: This is where your thermodynamic math plugs in. If your viscosity output (μ) is too high, the fluid drag drastically increases F_{load} . High viscosity means huge stretch.

Why PPRL matters?

Why this matters: Steel has a yield limit. If the Peak Load exceeds the tensile strength of the steel rod, the rod snaps, the pump falls to the bottom of the well, and Oil India has to spend thousands of dollars to fish it out. Your teammate uses this equation to prove their pump simulation won't break the equipment.

23

Tuesday

Week-09
Day 054-3112010
February

• Volumetric Displacement : $\rightarrow$

$$V = (\text{Area})_{\text{plunger}} \cdot (S - \Delta L) \cdot N$$

It calculates the final output, how much oil actually makes it to the surface per minute.

S = Surface stroke length (How far the motor pulls)

ΔL = Rod stretch Calculated in Egn (1).

N = Pumping speed (strokes per min)

February 2010

Monday	1	8	15	22
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Today's Priority

Need of Volumetric Displacement:

Why this matters: Notice that the stroke length (S) is reduced by the rod stretch (ΔL). If you (the Thermal Lead) do a poor job heating the reservoir, the viscosity stays high, the stretch becomes massive, and the volume (V) drops to near zero.

By running your thermal equations first, and feeding the resulting viscosity into these mechanical equations, you and your teammate create a complete, end-to-end Digital Twin.